Global Mortality Analysis

Insight Report, 1990–2019

NAME	Kaif Kadar Sayyed	**BATCH**	CX-134
COURSE	Business Analyst	**TOOL**	Tableau

1. Overview

This project analyses annual deaths by cause across countries from 1990 to 2019. It answers three questions: which causes of death dominate worldwide, how lifestyle diseases compare with infectious diseases and external causes, and where violence-related deaths are concentrated. The analysis is delivered as three Tableau dashboards built from ten worksheets.

2. Data and Method

- **Dataset:** annual number of deaths by cause, by country and year, 1990–2019.
- **Scope:** a calculated field, Is Country, removes aggregate rows (for example World and regional totals) so deaths are not counted twice. Every sheet uses it as a shared filter.
- **Grouping:** causes were grouped into three categories (Lifestyle, Infectious, External) with a calculated field.
- **Measures:** all figures are total deaths summed over the selected years, not annual averages.
- **Tool:** Tableau Desktop, using parameters, calculated fields, context filters, colour legends and dashboard filter actions.

3. Dashboard 1: Global Disease Burden

Key findings

1. **Cardiovascular diseases are the largest cause of death by a wide margin.** They account for about 447.7M deaths over the period, nearly double the second-ranked cause, neoplasms (cancers), at 229.8M.
2. **The top ten causes are led by chronic and respiratory conditions.** Chronic respiratory diseases (104.6M) and lower respiratory infections (83.8M) follow, then neonatal disorders (76.9M), diarrheal diseases (66.2M) and digestive diseases (65.6M).
3. **Communicable diseases remain in the top ten.** Tuberculosis (45.9M) and HIV/AIDS (36.4M) still rank, which shows the burden has not shifted entirely to chronic disease.
4. **Trends differ by disease.** Annual cardiovascular deaths rose from roughly 12M in the early 1990s to about 18M by 2019, and neoplasm deaths rose from roughly 6M to 10M. HIV/AIDS deaths peaked in the early 2000s and then declined.
5. **Cardiovascular deaths are concentrated in the most populous countries.** The map shows China and India with the highest totals.

4. Dashboard 2: Global Health Risk

Key findings

1. **Lifestyle diseases dominate the three categories.** They total 621.3M deaths, about 72% of the 857.7M across the three categories, against 173.8M for infectious diseases (about 20%) and 62.6M for external causes (about 7%).
2. **Lifestyle diseases cause roughly 3.6 times as many deaths as infectious diseases.** This is the clearest single message of the project for public health planning.
3. **Diabetes and cardiovascular deaths rise together across countries.** On the log-scale scatter plot, countries with higher diabetes deaths also have higher cardiovascular deaths, and the trend line slopes upward. The two conditions share risk factors and tend to co-occur.
4. **Infectious disease deaths are heavily concentrated.** India is the darkest country on the map at about 47M, far above all others, with Nigeria also standing out.

5. Dashboard 3: Global Violence and Conflict

Key findings

1. **Interpersonal violence drives most violence deaths.** In nine of the top ten countries, interpersonal violence makes up most of the bar.
2. **Violence deaths are highly concentrated.** Brazil leads at about 1.7M deaths over the period, followed by India and Russia at about 1.25M each, then China, Colombia, South Africa, the United States, Mexico, Rwanda and the Philippines.
3. **Rwanda is the exception.** Its total is mostly conflict deaths, and the trend chart shows conflict and terrorism deaths spiking sharply in 1994, from roughly 0.1M a year to about 0.55M. This is consistent with the 1994 Rwandan genocide, though that context comes from outside the dataset.
4. **Self-harm is the largest of the three violence-related causes.** At about 0.75–0.85M deaths a year it exceeds interpersonal violence (about 0.4M) and conflict (about 0.1M outside spike years), and it stays stable across the period. It is shown on the trend chart but is not part of the country ranking.
5. **Road injuries dwarf violence among external causes.** The external-causes area chart shows roughly 2M deaths a year, with road injuries making up the largest band at about 1.1–1.2M, more than conflict and interpersonal violence combined.

6. Key Takeaways

- Chronic disease, led by cardiovascular disease and cancer, accounts for most deaths and is still growing in absolute terms.
- Infectious diseases are a smaller but significant share, concentrated in a few countries.
- Violence deaths are geographically concentrated, mostly interpersonal rather than conflict-driven, with conflict creating rare, extreme spikes.
- Among external causes, road injuries are a larger and more steady burden than violence.

7. Recommendations

1. **Prioritise cardiovascular prevention and screening.** Blood pressure control, diet and tobacco programmes target the single largest cause of death, and the diabetes link suggests combined screening would be efficient.
2. **Target infectious disease programmes at high-burden countries.** Concentrated deaths mean focused funding in a few countries could cut the global total substantially.
3. **Treat interpersonal violence as a public health issue in the top-ranked countries.** Brazil, India, Russia, Colombia and South Africa account for much of the total, so local prevention programmes would have the most effect.
4. **Put road safety alongside violence in external-cause budgets.** Road injuries cause far more deaths than violence and respond to well-known measures such as enforcement and vehicle standards.
5. **Build early-warning monitoring for conflict.** Conflict deaths are low most years but can multiply in a single year, so planning should allow for sudden spikes.

8. Limitations

- **Totals, not rates.** Figures are absolute deaths, so populous countries such as India and China rank high partly because of population size. Rates per 100,000 people would allow fairer comparison.
- **Cumulative sums.** Totals add up 30 years, so they mix long-term trends with one-off events.
- **Category choices.** The Lifestyle, Infectious and External groupings are analyst-defined, and other groupings would give different shares.
- **Data quality varies.** Death recording and cause attribution differ between countries, and the data ends in 2019, before any later events.
- **Narrow violence view.** The country ranking covers conflict and interpersonal violence only, not self-harm.
- **Chart readings.** Some yearly trend values are read from chart positions and rounded, so they are approximate.